

Total No. of Questions : 10]

SEAT No. :

P3251

[Total No. of Pages : 2

[5670]-519

B.E. (Civil)

AIR POLLUTION AND CONTROL

(2015 Course) (Semester - II) (Elective - III) (401009D)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve Q.1 or Q.2,Q.3 or Q.4,Q.5 or Q.6,Q.7 or Q.8,Q.9 or Q.10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of logarithmic tables, slide rule, Mollier charts, electronic pocket calculator and steam tables are allowed.*
- 5) *Assume suitable data, if necessary.*

- Q1)** a) What are the zones of atmosphere? Explain Chemosphere. [6]
b) Explain stable and unstable atmosphere. [4]

OR

- Q2)** a) Explain Coning plume behaviour with the help of neat sketch. [6]
b) Explain with a neat sketch location of sampling ports and traverse points. [4]

- Q3)** a) Draw schematically a sampling train for stack sampling indicating the equipment required. [6]
b) Explain effects of indoor air pollutants with sources. [4]

OR

- Q4)** a) Write a short note on High volume sampler with sketch. [6]
b) Explain method of measurement of odour. [4]

- Q5)** a) What are the advantages and disadvantages of Bag filter? Draw a neat sketch of bag filter. [8]
b) What are the types of scrubbers? Explain the Venturi Scrubber with a neat sketch [8]

OR

P.T.O.

- Q6)** a) Explain working principle of a gravity settling chamber with a neat sketch and how the smallest size of particles removed can be found out? [8]
- b) What are the objectives of air pollution control equipment? Explain with example control of air pollution at source. [8]

- Q7)** a) What is land use planning? Explain its importance in controlling air pollution. [8]
- b) Explain Cost-Benefit Analysis (CBA) or Cost-Benefit Ratio in view of: Purposes of CBA, Process, Evaluation, and Cost-benefit ratio analysis with example. [8]

OR

- Q8)** a) What is the purpose of environmental (protection) ACT 1986? Explain the powers and duties of central and state pollution control boards. [8]
- b) Explain an important provision made in "The AIR (Prevention and Control of Pollution) ACT 1981". [8]

- Q9)** a) Write the environmental rules 1999 (siting of industries) as per the notification of Ministry of Environment and Forests. [9]
- b) Write note on Emission standards for stationary sources [9]

OR

- Q10)** a) Explain roles of different authorities in the EIA Process. [9]
- b) Write a short note on the following EIA methodologies: [9]
- i) Networks and ii) Matrix Method

